UTM UNIVERSITI TEKNOLOGI MALAYSIA	**FACULTY OF COMPUTING**		
	SUBJECT: PROBABILITY & STATISTICAL DATA ANALYSIS		
	ASSESSMENT: EXERCISE 2	**CODE:** SCST1223	**WEEK:** 2

EXERCISE 2

1. Below data shows student's body weight in kg.

41	60	72	72	65	64	48	90
60	75	53	48	63	49	58	39
39	75	55	62	53	59	58	38
72	62	60	59	68	60	70	60
56	80	65	85	71	45	70	55

Construct a frequency distribution tables (frequency, relative frequency and percent frequency) based on the above data.

2. Based on the grades of 20 students below;

D	C	E	B	B	B	A	D	B	C
B	E	C	B	C	B	B	D	B	C

Construct a frequency distribution tables (frequency, relative frequency and percent frequency) based on the above data.

3. 30 students were asked for their course they major in. The responses were recorded as follow:

A	M	M	A	M	E
M	S	A	E	E	M
A	S	E	M	A	M
A	M	A	S	A	M
E	E	M	A	M	M

(M=Management; A=Accounting; E=Economics; S=Statistics)

a) What is the **frequency** for Management students?
b) What is the **relative frequency** for Accounting students?
c) What is the **percent frequency** for Statistic students?

4. Based on the test score below:
   74, 83, 69, 95, 78, 85, 42, 98, 73, 68, 90, 85, 84, 71, 88, 52, 94

   a) Develop frequency distribution table that consist of frequency, relative frequency and cumulative frequency.
   b) Plot histogram.

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5. Based on the data below:
   5, 6, 12, 13, 15, 18, 22, 50

   Find
   a) 5 summary data
   b) Construct skeletal / normal boxplot
   c) Construct modified boxplot

6. Based on the data below:
   300, 400, 450, 475, 500, 550, 650, 700, 700, 750

   Find
   a) 5 summary data
   b) Construct skeletal / normal boxplot
   c) Construct modified boxplot

MIDTERM TEST 2018/2019

A student representative studied the amount of money (in RM), School of Computing (SC) students spend on online game for a week, and the result is as follows:

23	60	79	32	57	74	52	70	82	36
80	77	81	95	41	65	92	85	55	76
52	10	64	75	78	25	80	98	81	67
41	71	83	54	64	72	88	62	74	43
60	78	89	76	84	48	84	90	15	79
34	67	17	82	69	74	63	80	85	61

(a) Construct a frequency distribution table as in Table 2 for the grouped data. (12 marks)
(Hint: Use 10 class intervals, beginning with lower class limit of 9 and class width of 9).

Table 2

Class	**Frequency**	**Cumulative Frequency**	**Relative Frequency**	**Cumulative Relative Frequency**
9 - 17				
…				
…				

(b) Draw a histogram to show the frequency distribution for your answer in (a). (3 marks)
(c) Plot another graph (a relative ogive) based on your answer in (a). (3 marks)
(d) Do you think SC students spend too much money on online games? Why? (2 marks)

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MIDTERM TEST 2016/2017

a) Air pollutant index (API) is used as reference to measure the level of air pollution. Table 1 shows the reading of air pollutant index in 19 areas taken at 12 noon. The API readings were grouped into according index.

Table 1: Air pollutant index in 19 areas taken at 12 noon

API	Frequencies
0-50 (good)	1
50-100 (moderate)	2
100-200 (Unhealthy)	6
200-300 (Very Unhealthy)	7
300-500 (Hazardous)	3

i) Sketch a histogram to represent the frequencies of API in table 1. (3 marks)

ii) In order to better understand the air pollutant situation, calculate relative frequencies (2 decimal places) and state the percentage of areas affected with API ≥ 100. (2 marks)

iii) Assume that an API reading of 950 was taken at 12 noon at one of the areas. Does this value affect the mean value for API? Justify your answer. (2 marks)

MIDTERM TEST 2019/2020

The temperatures (in ° C) measured from 30 Covid-19 patients on day 5 of quarantine in isolation ward are listed below:

35.5	35.7	35.8	35.9	36.1	36.1	36.3	36.4	36.5	36.6
36.7	36.7	36.7	36.9	37.0	37.0	37.0	37.1	37.2	37.2
37.4	37.5	37.7	37.7	37.8	38.0	38.1	38.1	38.3	38.7

(a) Draw stem-and-leaf for the above data.

(b) Show the calculations for the 1st, 2nd, 3rd quartiles and the inter-quartile range.

(c) Draw the boxplot of these data.

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